CHAPTER 12 TRANSFER OF THERMAL ENERGY

2.3 Transfer of thermal energy

2.3.1 Conduction

1 Describe experiments to distinguish between good and bad thermal conductors

2 Describe thermal conduction in all solids in terms of atomic or molecular lattice vibrations and also in terms of the movement of free (delocalised) electrons in metallic conductors

2.3.2 Convection

1 Explain convection in liquids and gases in terms of density changes and describe experiments to illustrate convection

2.3.3 Radiation

1 Describe the process of thermal energy transfer by infrared radiation and know that it does not require a medium

2 Describe the effect of surface colour (black or white) and texture (dull or shiny) on the emission, absorption and reflection of infrared radiation

3 Describe how the rate of emission of radiation depends on the surface temperature and surface area of an object

4 Describe experiments to distinguish between good and bad emitters of infrared radiation

5 Describe experiments to distinguish between good and bad absorbers of infrared radiation

2.3.4 Consequences of thermal energy transfer

1 Explain everyday applications using ideas about conduction, convection and radiation, including:

(a) heating objects such as kitchen pans

(b) heating a room by convection

(c) measuring temperature using an infrared thermometer

(d) using thermal insulation to maintain the temperature of a liquid and to reduce thermal energy transfer in buildings

1. O/N/23/11/Q16

A student on a camping expedition cools a sealed bottle of water which is at the same temperature as the surrounding air.

Which method cools the water at the greatest rate?

A Wrap the bottle in aluminium foil and place it in a shady place.

B Wrap the bottle in dry, white paper and put it in a sunny place.

C Wrap the bottle in foam and put it in a breeze.

D Wrap the bottle in wet paper and put it in a breeze.

2. O/N/23/11/Q17

A student suggests three factors that affect the rate of emission of thermal energy by radiation from a hot object. These are:

1 the surface temperature of the object

2 the surface area of the object

3 the surface colour of the object.

Which suggestions are correct?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

3. O/N/23/12/Q21

Solid metals are good thermal conductors.

How closely packed are their particles and do the metals contain free electrons?

	packing of particles	contains free electrons
A	close together	no
B	close together	yes
C	far apart	no
D	far apart	yes

4. O/N/23/12/Q22

Cotton wool is a good thermal insulator.

It consists of many threads of cotton which are tangled together to make a material which can easily be pulled apart or compressed. The diagram shows the structure of cotton wool.

Why is cotton wool a good thermal insulator?

A Each thread is very thin.

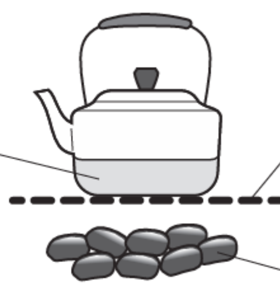
B It has a large surface area because it is made of many threads.

C It has a low density because of the trapped air.

D It traps air and prevents convection.

5. O/N/23/12/Q23

A metal kettle is heated on a grid above red-hot glowing coal.

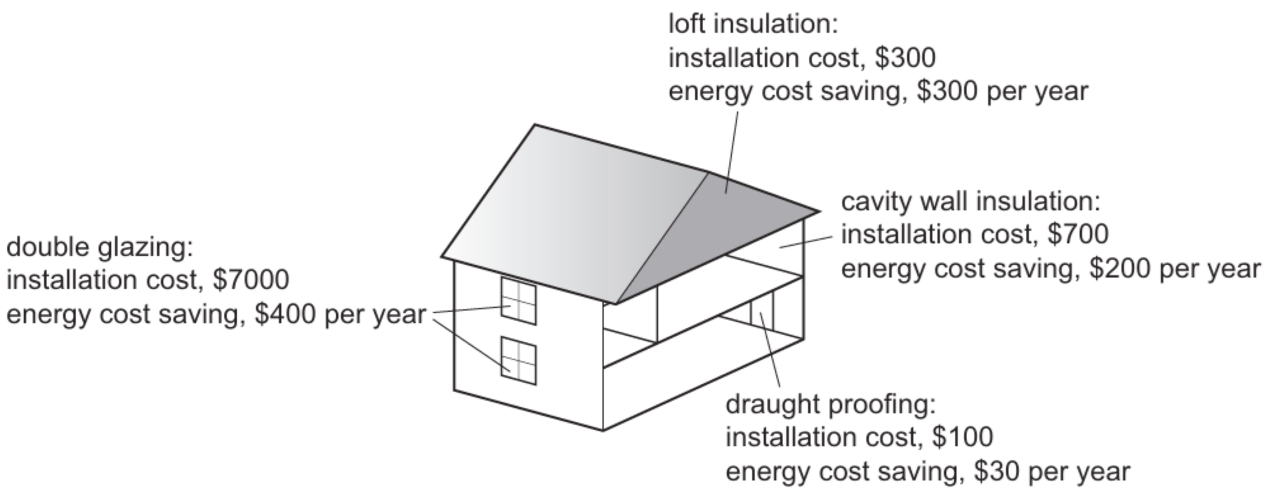


What are the main thermal energy transfers through the bottom of the kettle and within the water?

	through kettle	within water
A	conduction	conduction
B	radiation	convection
C	conduction	convection
D	radiation	conduction

6. M/J/23/11/Q18

A house is to be thermally insulated to reduce thermal energy loss and reduce fuel costs.



Which type of thermal insulation pays back the cost of installation in the shortest time?

A cavity wall insulation

B double glazing

C draught proofing

D loft insulation

7. M/J/23/12/Q17

Which statement about copper explains why it is a better thermal conductor than glass?

A Atomic vibration is passed on to neighbouring copper atoms slowly.

B Atoms move through the copper and pass on kinetic energy.

C There are density changes within the copper.

D There are free electrons in the copper.

8. M/J/23/12/Q18

Which statement describes the transfer of thermal energy in a liquid by convection?

A A decrease in its density causes the heated liquid to rise.

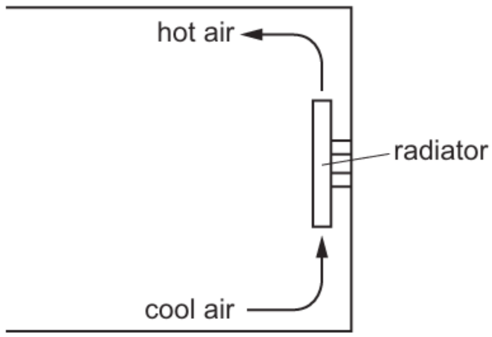
B Free electrons carry energy large distances through the liquid.

C Infrared radiation passes through the liquid and transfers energy.

D Particles of the liquid vibrate and pass energy to neighbouring particles.

9. O/N/22/11/Q17

The diagram shows a radiator heating the air in a room.



What is the name of this process?

A conduction

B convection

C evaporation

D expansion

10. O/N/22/11/O18

What is a process of heat transfer that can take place in a vacuum?

A conduction

B convection

C evaporation

D radiation

11. M/J/22/11/Q13

A building has walls made from wood and steel.

What describes the way in which thermal energy is transferred through the walls?

A conduction through the wood by movement of molecules through the wood only

B conduction through the metal by both the vibration of particles and movement of electrons through the steel

C conduction through the wood by both the vibration and movement of molecules through the wood

D convection through the metal by movement of electrons through the steel only

12. M/J/22/12/Q16

How is heat energy transferred through the vacuum of space from the Sun to the Moon's surface?

A conduction only

B convection only

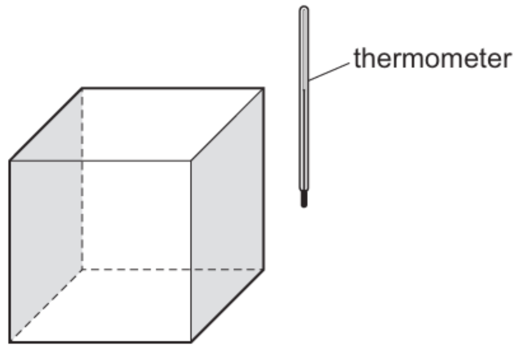
C radiation only

D conduction, convection and radiation

13. M/J/22/12/Q17

A metal cube contains boiling water.

Each vertical face of the cube is painted a different colour or has a different texture.



Identical thermometers are held at equal distances from each vertical face.

Near which face does a thermometer read the greatest temperature?

A dull black

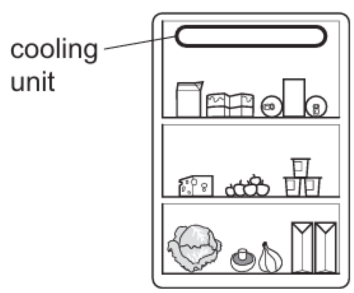
B dull white

C shiny black

D shiny white

14. O/N/21/11/Q12

The diagram shows the inside of a refrigerator.



When the refrigerator is first switched on, what happens to the air near the cooling unit?

	the molecules of this air	the density of this air
A	become smaller	decreases
B	become smaller	increases
C	move closer together	decreases
D	move closer together	increases

15. O/N/21/12/Q15

What is the colour and what is the texture of a good absorber of infrared radiation?

A black and shiny

B black and dull

C white and shiny

D white and dull

16. M/J/21/11/Q15

A copper rod is heated at one end.

Which statement describes how heat transfer occurs in the copper?

A Energetic copper molecules move from the cooler end to the hotter end.

B Energetic copper molecules move from the hotter end to the cooler end.

C Energetic free electrons move from the cooler end to the hotter end.

D Energetic free electrons move from the hotter end to the cooler end.

17. M/J/21/12/Q18

Which heat transfer processes do **not** require a medium?

A conduction only

B convection only

C radiation only

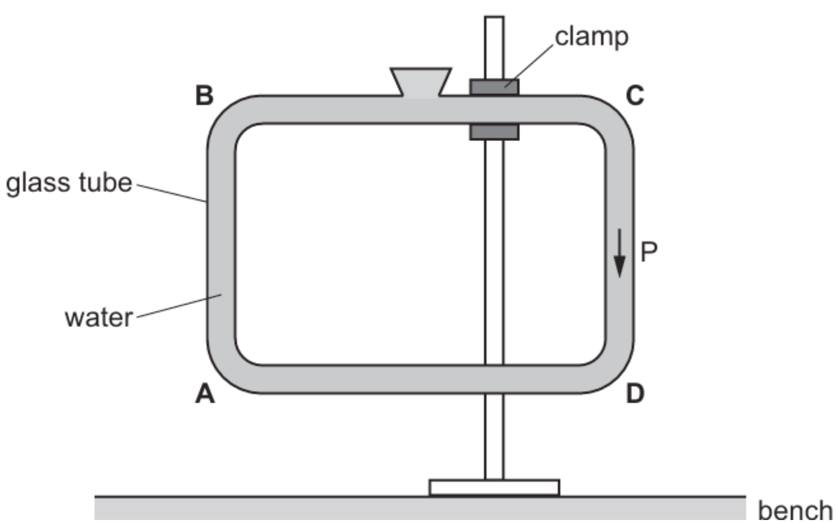
D conduction and radiation

18. O/N/20/12/Q17

The diagram shows a glass tube filled with water and suspended above a bench. The water is free to circulate around the tube.

At point P, there is a convection current moving in a downwards direction.

At which point is the tube heated to cause this convection current?



19. M/J/20/11/Q21

Which description of a dull black surface is correct?

A good emitter, good absorber and good reflector of radiation

B good emitter, poor absorber and poor reflector of radiation

C good emitter, good absorber and poor reflector of radiation

D poor emitter, poor absorber and poor reflector of radiation

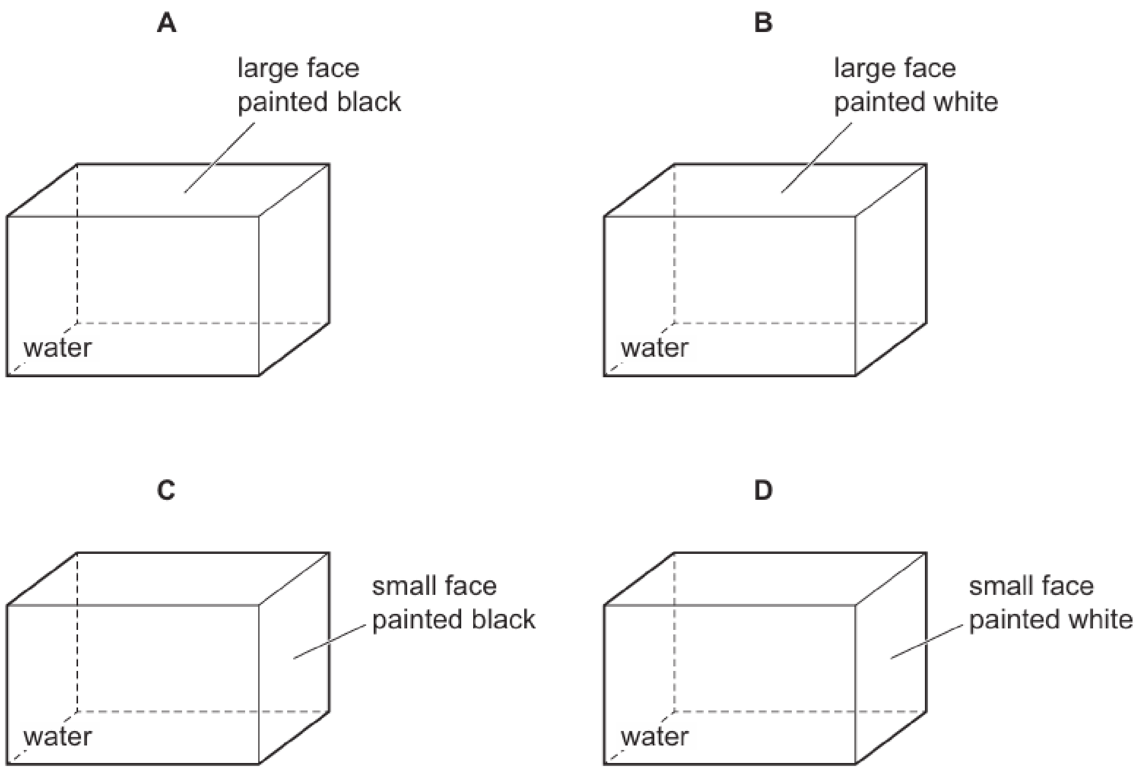
20. M/J/20/12/Q23

Four metal containers, with identical dimensions, are filled with water at 90 °C.

All the faces, except one, of each container are covered in a very good insulator.

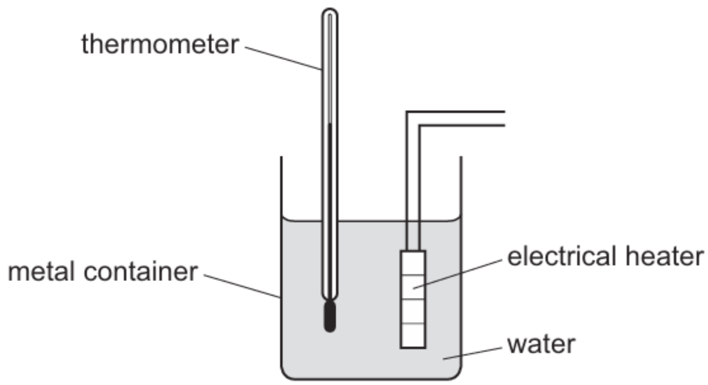
The one exposed face on each container is painted either black or white.

In which container does the water cool the fastest?



21. M/J/19/11/Q17

The diagram shows a set of apparatus used to determine the specific heat capacity of water.



What does **not** affect the rate at which energy is lost to the surroundings?

A insulating the container

B placing a lid on the container

C polishing the outer surface of the container

D moving the thermometer closer to the heater

22. O/N/18/11/Q17

A cold solid is placed on top of a hot solid. Thermal energy flows from the hot solid to the cold one.

What is the explanation for this?

A A hot solid expands, so its molecules will move further apart.

B Energy is passed from one molecule to the next.

C Heat always rises.

D Molecules are free to move randomly through the solids.

23. O/N/18/12/Q17

A cold bottle containing a drink is placed on a table on a warm day. Drops of water form on the outside of the bottle.

Which process causes the drops to form?

A condensation

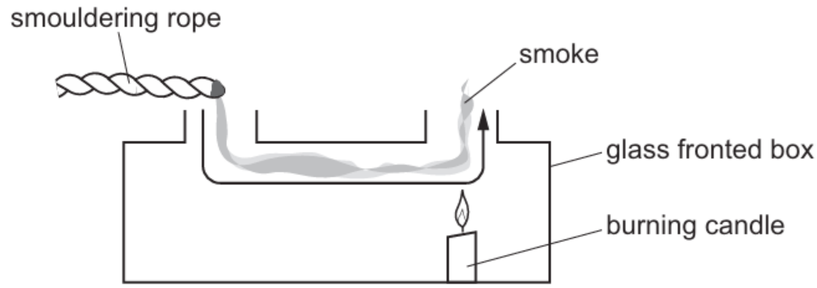
B conduction

C convection

D evaporation

24. M/J/18/11/Q20

When a piece of smouldering rope is held at the opening of the box in the diagram, smoke moves in the direction indicated.



What is responsible for the movement of the smoke?

A convection

B movement of free electrons

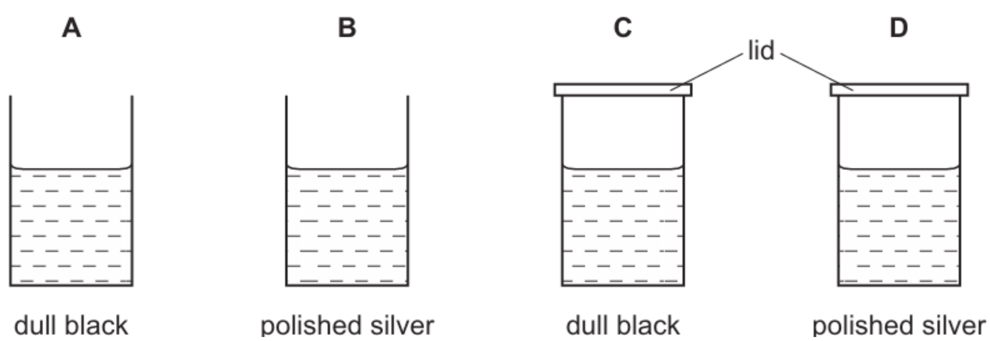
C radiation

D vibration of molecules

25. M/J/18/11/Q21

The diagrams show four identical cans with their outside surfaces either polished silver or painted dull black. Each can contains the same volume of water, initially at 80 °C.

After five minutes in a cool room, which can contains the coolest water?



26. M/J/18/12/Q24

Samples of four different materials at room temperature are heated from below and heat is transferred upwards.

In which material is all of the heat transferred by the vibration of molecules?

A air

B mercury

C rubber

D water

27. O/N/17/11/Q16

In which substance is the conduction of thermal energy mainly due to the movement of electrons?

A air

B ice

C iron

D water

28. O/N/17/11/Q17

Four similar metal plates are the same distance from a heater that emits infra-red radiation.

The plates are painted dull black, dull white, shiny black and shiny white.

Which plate absorbs the most radiation and which plate reflects the most radiation?

	absorbs most radiation	reflects most radiation
A	dull black	dull white
B	dull black	shiny white
C	shiny black	dull black
D	shiny white	dull black

29. O/N/17/12/Q18

A heater is designed to radiate thermal energy.

Which change to the design decreases the thermal energy emitted by radiation?

A a darker coloured surface

B a higher surface temperature

C a larger surface area

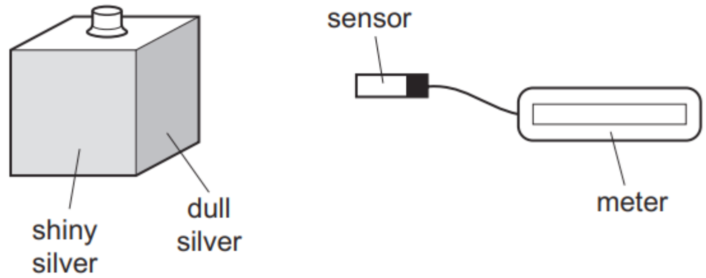
D a shinier surface

30. M/J/17/12/Q25

A metal box has four different surfaces; dull black, shiny black, dull silver and shiny silver.

The box is filled with boiling water so that each surface is at the same temperature.

A sensor measures the amount of radiation from each surface.



Which surface emits the least radiation and which surface emits the most radiation?

	least	most
A	dull black	shiny silver
B	dull silver	shiny black
C	shiny black	dull silver
D	shiny silver	dull black

31. O/N/16/11/Q20

Which statement about the transfer of thermal energy is correct?

A Transfer by radiation does not require a medium.

B Transfer is from a region of lower temperature to one of higher temperature.

C Transfer upwards in fluids is mainly through the vibrations of neighbouring particles.

D Transfer in solids is by means of density changes in the material.

32. O/N/16/12/Q24

In which states of matter does convection occur?

A gases, liquids and solids

B gases and liquids only

C gases and solids only

D liquids and solids only

33. M/J/16/11/Q18

Which statement about thermal radiation is correct?

A In a vacuum, thermal radiation travels at the speed of light.

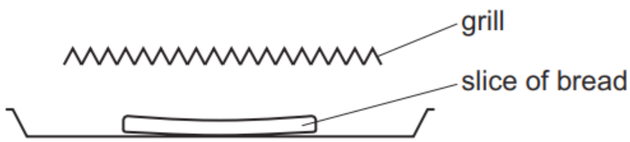
B Thermal radiation is a longitudinal wave.

C Thermal radiation travels as an ultra-violet wave.

D White surfaces are better emitters of thermal radiation than black surfaces.

34. M/J/16/12/Q15

A slice of bread is placed under a red-hot electric grill to make toast.



How does heat energy reach the bread?

A conduction and convection

B conduction only

C convection and radiation

D radiation only

35. O/N/15/12/Q20

Which statement is correct?

A Infra-red radiation cannot travel in a vacuum.

B Infra-red radiation cannot travel in solids or in gases.

C Infra-red radiation can only travel in a vacuum.

D Infra-red radiation can travel in a vacuum and in gases.

36. M/J/15/11/Q15

A solid bar is heated at one end.

How is thermal energy transferred to the other end of the bar?

A Heated molecules move along the bar, carrying energy to the other end.

B Heated molecules move along the bar, giving energy to others along the bar.

C Heated molecules stay completely still, but give energy to other molecules.

D Heated molecules vibrate more rapidly and pass energy to other molecules.

37. M/J/15/11/Q16

The tubes inside solar heating panels use the Sun's radiation to warm water.

Why are the tubes painted black?

A Black surfaces absorb radiation well.

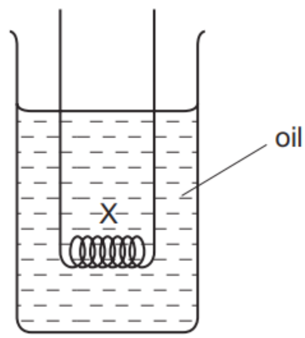
B Black surfaces conduct heat well.

C Black surfaces emit radiation well.

D Black surfaces reflect radiation well.

38. M/J/15/12/Q15

An electrical heater is placed in a beaker of cold oil, as shown.



The heater is switched on.

What happens to the liquid at X?

A It becomes less dense and falls.

B It becomes less dense and rises.

C It becomes more dense and falls.

D It becomes more dense and rises.

39. O/N/14/12/Q11

Which statement about copper explains why it is a better conductor of heat than glass?

A Atomic vibration is passed on to neighbouring copper atoms quickly.

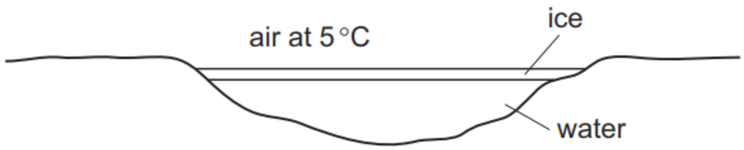
B Atoms move through the copper and pass on kinetic energy.

C There are density changes within the copper.

D There are free electrons in the copper.

40. O/N/14/12/Q12

The diagram shows a frozen pond with the surface of the ice slowly melting as heat is transferred from the warmer air above it.



By which processes is heat transferred from the air to the ice?

A conduction, convection and radiation

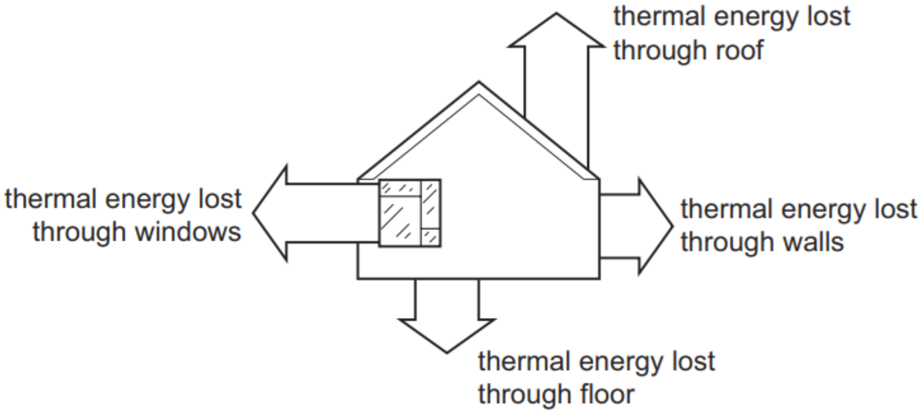
B conduction and convection only

C convection and radiation only

D radiation and conduction only

41. O/N/13/11/Q13

On a cold afternoon, a house loses 54 MJ of thermal energy (heat) to its surroundings as shown.



The heating system must supply more than 54 MJ of thermal energy to keep the temperature of the house constant.

Which statement explains this?

A The extra thermal energy is lost from the house to the surroundings by other means.

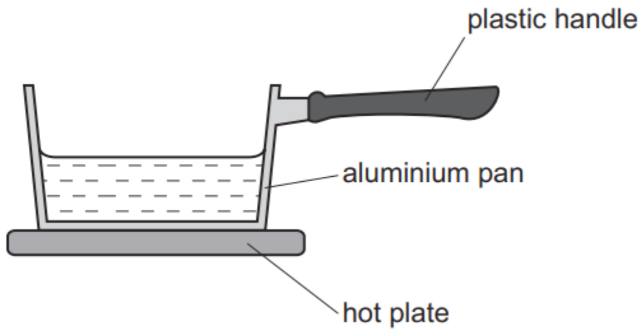
B The extra thermal energy keeps the house warmer than the surroundings.

C The temperature of the surroundings decreases continuously during this period.

D The thermal insulation of the roof is extremely ineffective.

42. O/N/13/12/Q15

A saucepan is used to heat up some water.



How is heat transferred through the aluminium pan and through the plastic handle?

	heat is transferred through the aluminium pan by	heat is transferred through the plastic handle by
A	the movement of free electrons and the vibration of atoms	the movement of free electrons and the vibration of molecules
B	the movement of free electrons and the vibration of atoms	the vibration of molecules only
C	the movement of free electrons only	the movement of free electrons and the vibration of molecules
D	the movement of free electrons only	the vibration of molecules only

43. O/N/13/12/Q16

A silver cup is filled with boiling water from a kettle.

A man touches the outside surface of the cup and finds that it is extremely hot.

Why is the surface so hot?

A Convection takes place in the boiling water.

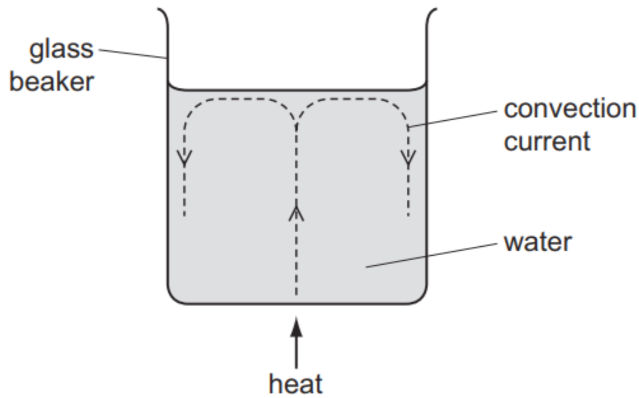
B Silver is a good conductor of heat.

C The boiling water gives out latent heat.

D The shiny surface is a good emitter of infra-red radiation.

44. M/J/13/11/Q13

A glass beaker contains water. When the centre of the base of the beaker is heated, a convection current is set up.



Which statement explains this?

A The evaporation of water causes water molecules to rise to the surface.

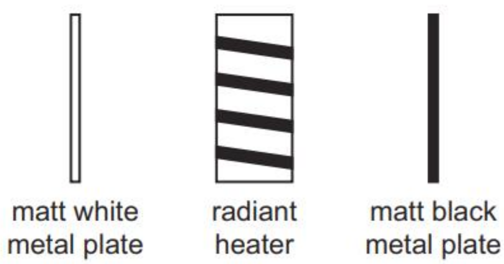
B The expansion of water molecules causes them to rise to the surface.

C The water above the heat source rises because it becomes less dense.

D The water at the sides sinks because it becomes less dense.

45. M/J/13/11/Q14

Two identical metal plates are painted, one matt (dull) white and the other matt black. These are placed at equal distances from a radiant heater as shown. The heater is turned on for five minutes.

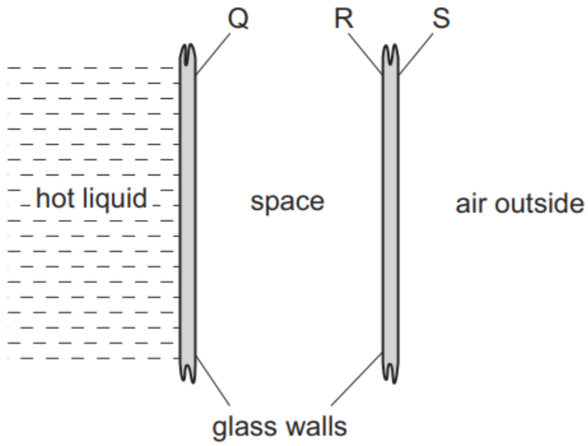


Which metal plate absorbs more energy and which plate emits more energy in this time?

	absorbs more	emits more
A	black	black
B	black	white
C	white	black
D	white	white

46. O/N/12/11/Q17

A student uses a double-walled glass vessel to contain a hot liquid.



What reduces the heat loss by radiation?

A a vacuum in the space between the walls

B painting surface Q black

C painting surface R black

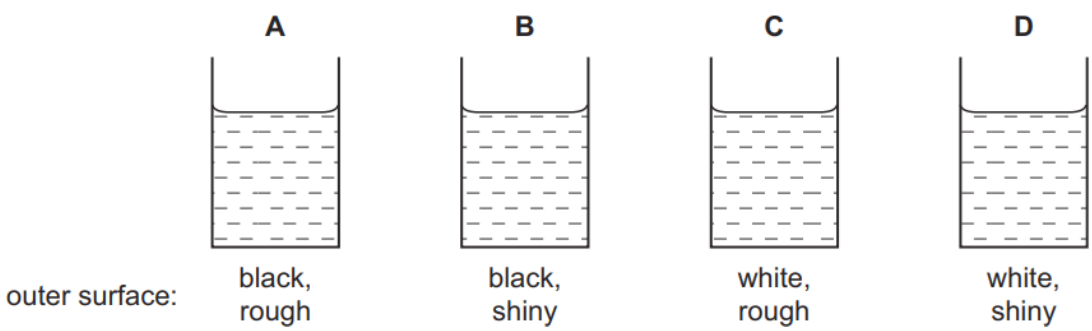
D painting surface S silver

47. O/N/12/12/Q17

Four metal cans are identical except for the colour and the texture of their outer surfaces.

100 cm^3 of water at 70 °C is poured into each can.

Which cools the most rapidly?



48. M/J/12/11/Q17

A rod of metal is heated at one end.

Which statement best describes the conduction of heat through the metal?

A Atoms move from the hot end and hit electrons at the cold end.

B Atoms vibrate and hit atoms at the cold end.

C Free electrons move from the hot end and hit atoms further along the rod.

D Free electrons vibrate and pass energy to free electrons further along the rod.

49. M/J/12/12/Q17

How is heat conducted in a metal?

A by movement of electrons through the metal only

B by movement of atoms through the metal only

C by vibration of atoms and movement of electrons through the metal

D by vibration of atoms only

50. O/N/11/11/Q15

How does heat transfer through a vacuum take place?

A by conduction, convection and radiation

B by conduction only

C by convection only

D by radiation only

51. M/J/11/11/Q16

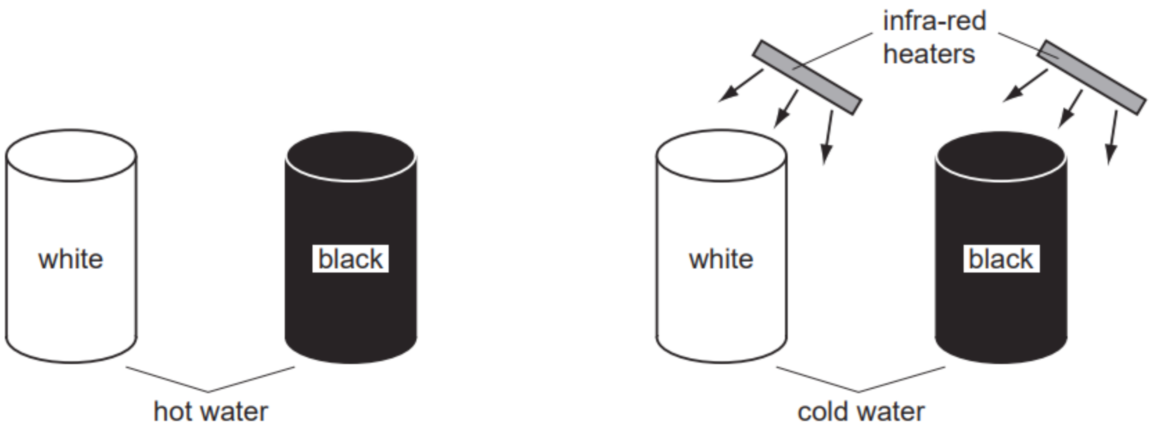
A balloon filled with air is gently heated.

What happens to the mass and the density of the air inside the balloon?

	mass	density
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

52. O/N/09/Q14

The diagrams show four cans in a cool room. They are painted as shown. One pair is filled with hot water and left to cool down. The other pair is filled with cold water and placed near infra-red heaters.



The hot water in the black can cools more quickly than the hot water in the white can. The cold water in the black can heats up more quickly than the cold water in the white can.

Which row shows the reasons for this?

	better emitter of infra-red	better absorber of infra-red
A	black	black
B	black	white
C	white	black
D	white	white

53. M/J/08/Q14

In a vacuum flask, which methods of heat transfer are prevented by the vacuum?

A conduction only

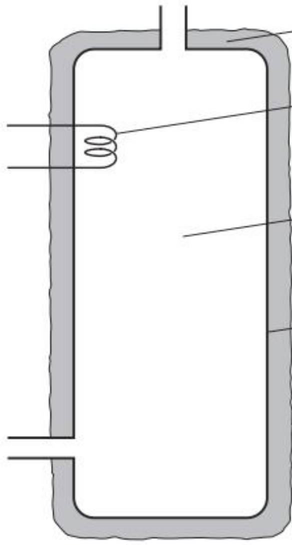
B convection only

C conduction and convection only

D conduction. convection. and radiation

54. O/N/07/Q18

Water at the top of a hot-water tank is heated and the water becomes hot. No water enters or leaves the tank.



Water at the bottom of the tank stays cold for some time.

Why is this?

A Cold water at the top of the tank falls to the bottom.

B Hot water at the bottom of the tank rises to the top.

C Water is a poor conductor of heat.

D The insulation is a poor conductor of heat.

55. O/N/06/Q18

The heat from the hot water in a metal radiator passes through the metal and then spreads around the room.

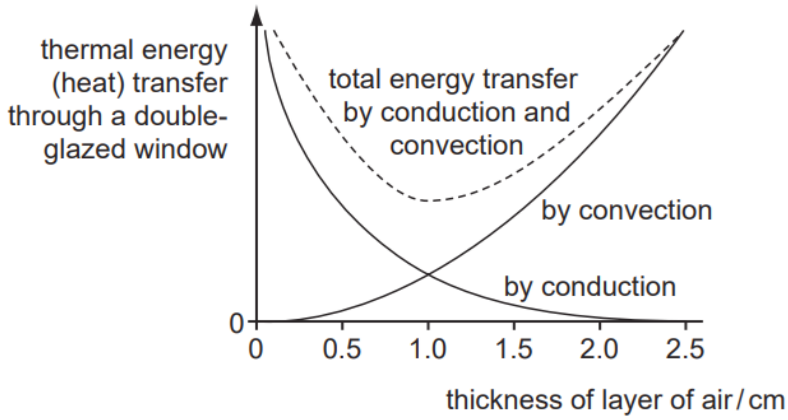
What are the main processes by which the heat is transferred through the radiator and then spread around the room?

	through the metal radiator	around the room
A	conduction	conduction
B	conduction	convection
C	radiation	conduction
D	radiation	convection

56. M/J/07/Q18

A double-glazed window has two sheets of glass separated by a layer of air.

Thermal energy is conducted and convected through the layer of air. The amount of conduction and convection varies with the thickness of the layer of air, as shown in the graph.



Which thickness of air produces the smallest energy transfer, and why?

A 0.5 cm because there is little convection

B 1.0 cm because the total thermal energy transfer is least

C 1.5 cm because the total thermal energy transfer is small and conduction is low

D 2.0 cm because there is little conduction

57. M/J/06/Q20

Density changes are responsible for which method of thermal energy transfer?

A conduction only

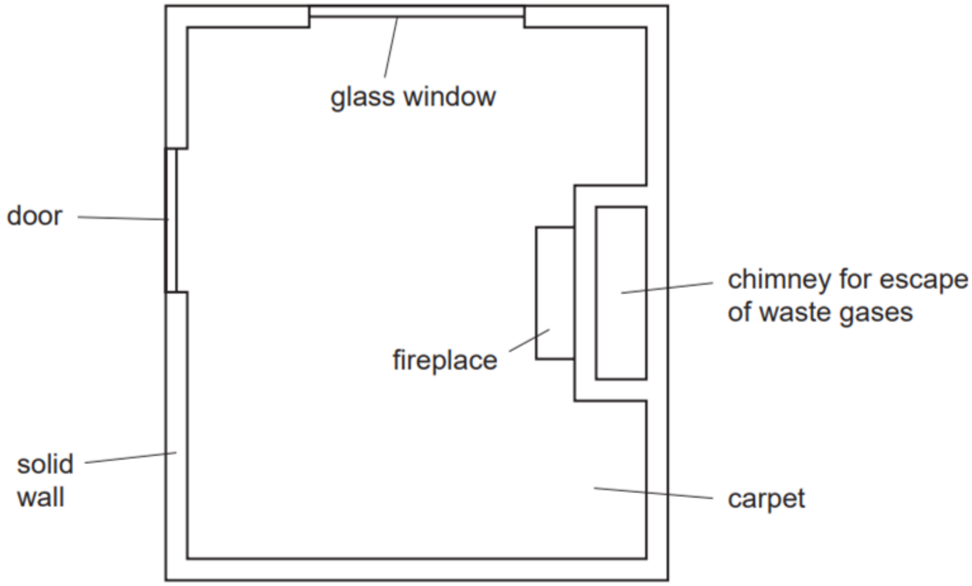
B convection only

C radiation only

D conduction, convection and radiation

58. O/N/05/Q18

The diagram shows a room seen from above. It is cold outside the room. The room is heated by a small fire in the fireplace.



Where is most heat lost by convection?

A carpet

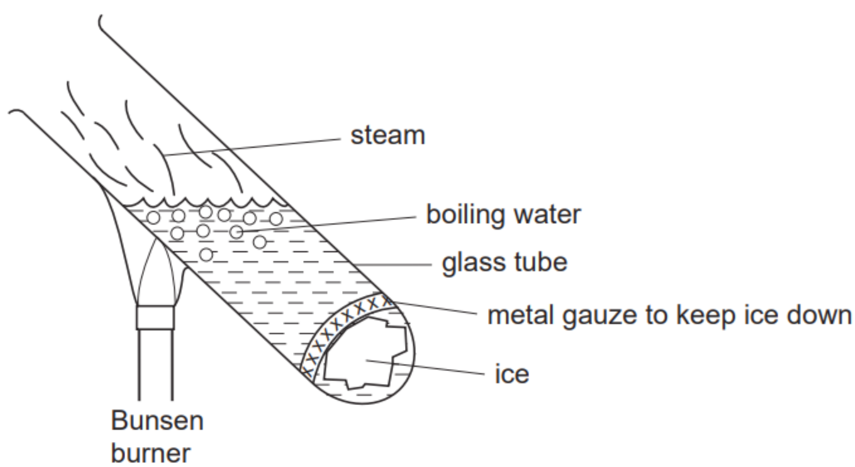
B chimney

C glass window

D solid wall

59. O/N/05/Q19

An experiment is carried out as shown in the diagram.



Why does the ice take a long time to melt, even though the water at the top of the tube is boiling?

A Convection never occurs in water.

B Ice is a poor conductor of heat.

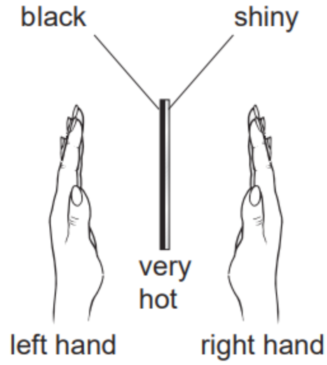
C The gauze prevents the energy reaching the ice.

D Water is a poor conductor of heat.

60. M/J/05/Q18

The diagram shows a thick copper plate that is very hot. One side is black, the other is shiny.

A student places her hands the same distance from each side as shown.



Her left hand feels warmer than her right hand.

Which statement is the correct conclusion from the experiment?

A The black side is hotter than the shiny one.

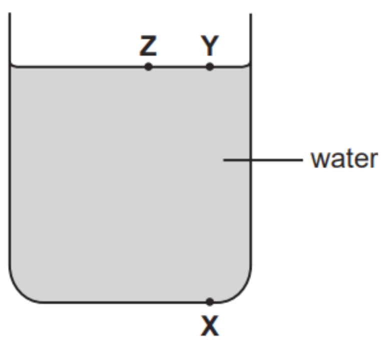
B The black side radiates more heat.

C The shiny side radiates more heat.

D The shiny side is cooling down faster than the black side.

61. O/N/04/Q21

A teacher has a large tank of water in which he wants to set up a convection current.



Which of the following arrangements would do this?

A cooling at **X**

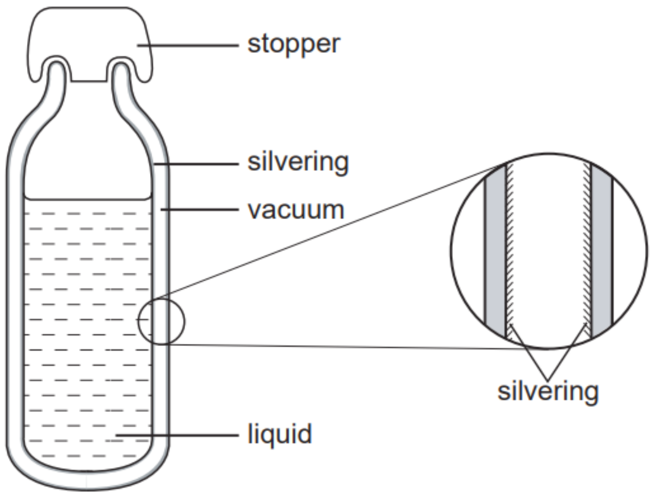
B cooling at **Y**

C heating at **Y**

D heating at **Z**

62. M/J/04/Q18

The diagram shows a vacuum flask and an enlarged view of a section through the flask wall.



The main reason for the silvering is to reduce heat transfer by

A conduction only.

B radiation only.

C conduction and convection.

D convection and radiation.

ANSWERS

1. D
2. A
3. B
4. D
5. C
6. D
7. D
8. A
9. B
10. D
11. B
12. C
13. A
14. D
15. B
16. D
17. C
18. A
19. C
20. A
21. D
22. B
23. A
24. A
25. A
26. C
27. C
28. B
29. D
30. D
31. A
32. B
33. A
34. D
35. D
36. D
37. A
38. B
39. D
40. D
41. A
42. B
43. B
44. C
45. A
46. D
47. A
48. C
49. C
50. D
51. C
52. A
53. C
54. C
55. B
56. B
57. B
58. B
59. D
60. B
61. B
62. B

1. O/N/23/21/Q3

When the temperature of a liquid increases, the kinetic energy of its particles increases and the liquid expands.

(a) Explain, by referring to particles, why a liquid expands when heated.

....

....

....

.... [2]

(b) The heater in a kettle is near to the base of the kettle. The kettle is filled with water at a temperature of 17 °C and the heater is switched on.

(i) State the name of the method of thermal energy transfer that causes energy to be transferred to the water that is touching the heater.

.... [1]

(ii) Explain how thermal energy is transferred through all of the water in the kettle.

....

....

....

.... [3]

(iii) State the boiling temperature of water at standard atmospheric pressure. Include the unit in your answer.

.... [1]

(iv) The kettle contains 2.5 kg of water at standard atmospheric pressure.

The specific heat capacity of water is 4200 J / (kg °C).

When the water reaches its boiling temperature, the kettle switches off.

Calculate the increase in the internal energy of the water.

increase in internal energy = ... J [3]

2. O/N/22/22/Q3

Fig. 3.1 shows a man standing underneath an outdoor heater on a cold evening.

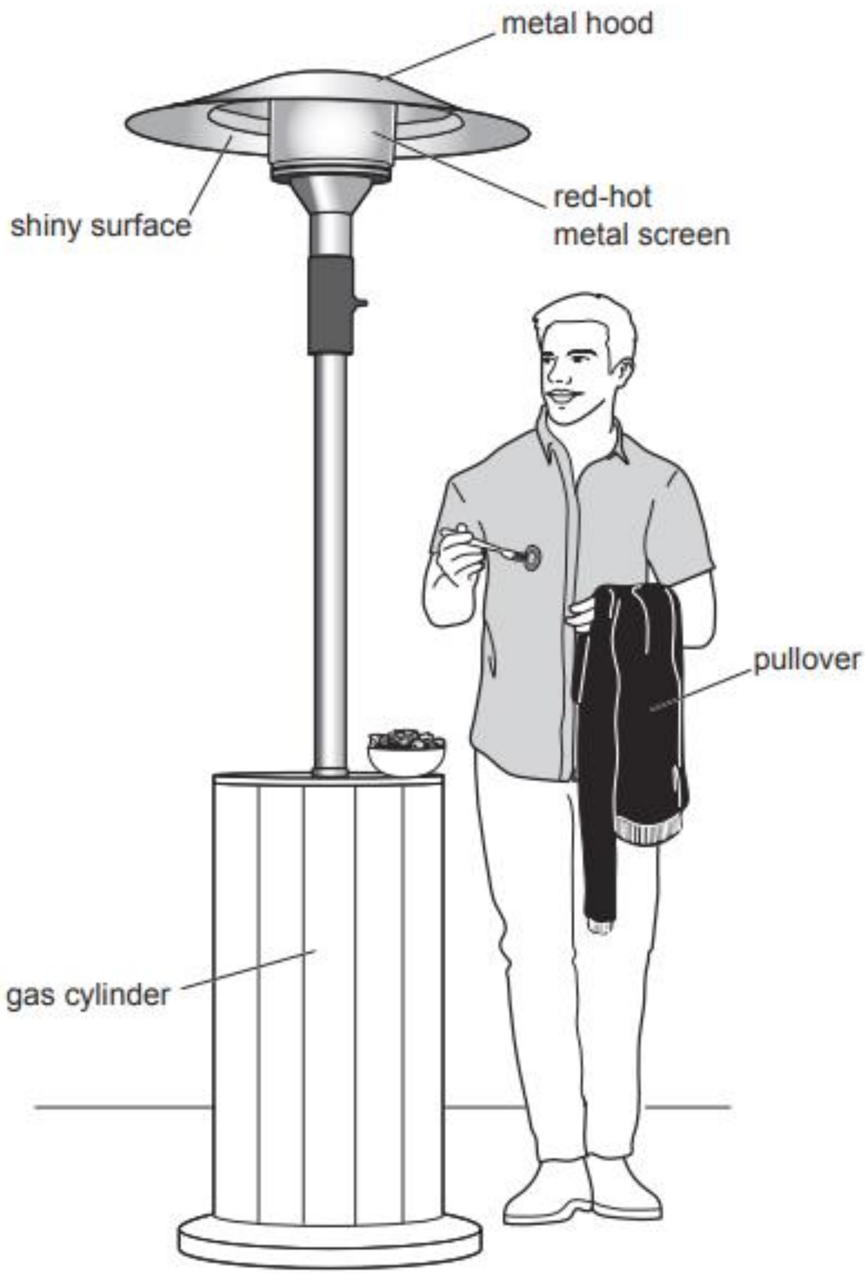


Fig. 3.1

Gas in the cylinder at the base of the heater is the fuel for the heater. When the heater is operating, the gas travels to the top of the heater where it burns.

(a) State the form of energy stored in the gas that is transferred by the heater.

.... [1]

(b) A metal screen surrounding the burning gas is heated by the burning gas until it is red-hot. The hot metal screen warms the man who is standing underneath it.

(i) Describe how thermal energy in the red-hot metal screen is transferred to the man and how it warms him.

....

....

....

....

.... [3]

(ii) At the top of the heater is a metal hood that has a shiny lower surface.

Explain why this makes the energy transfer from the metal screen more efficient.

....

....

.... [2]

(iii) The air temperature decreases and the man puts on a black pullover.

Explain **one** way in which wearing the black pullover helps to keep the man warm.

....

....

.... [2]

[Total: 8]

3. M/J/21/21/Q2

Fig. 2.1 shows a hollow metal cube filled with boiling water. The temperature of the four vertical surfaces are equal but each surface has a different colour or texture.

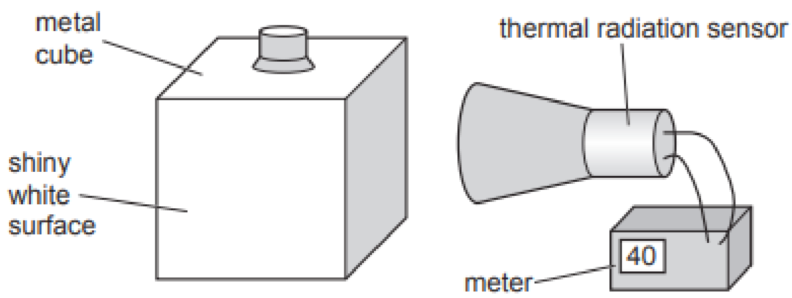


Fig. 2.1

A thermal radiation sensor is placed the same distance from each surface and the meter reading measures the thermal radiation emitted from each surface.

(a) The results are shown in **Fig. 2.2**.

Draw a line linking each type of surface with the appropriate meter reading. One line has been drawn for you.

type of surface	meter reading
dull black	40
dull white	60
shiny black	80
shiny white	100

Fig. 2.2

[2]

(b) The hot surfaces produce a convection current in the air outside the metal cube.

Describe how the convection current is produced.

....

....

....

....

.... [3]

4.O/N/20/22/Q3

Fig. 3.1 shows a hot water tank that contains two electric heaters X and Y.

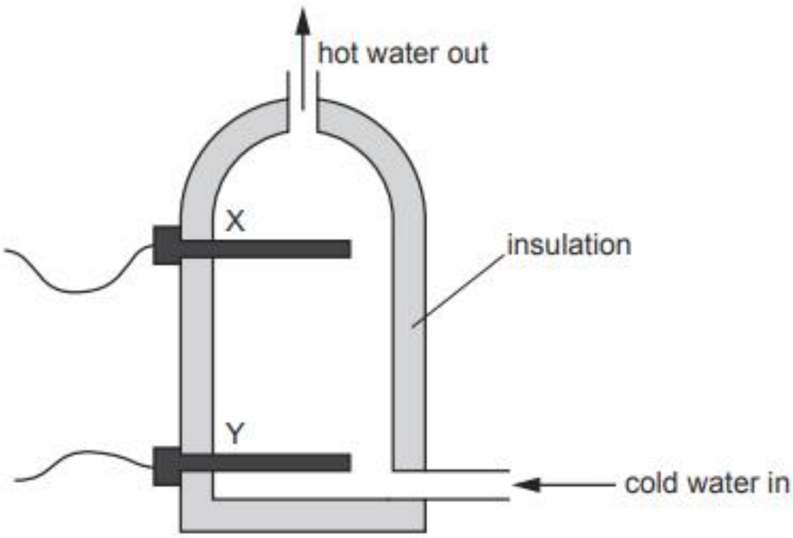


Fig. 3.1

Heater X is used during the daytime but heater Y is only used at night when electricity is cheaper.

(a) The tank is full of cold water and X is switched on.

The temperature of **all** the water above X increases very quickly but the temperature of the water below X increases much more slowly.

(i) Explain the process that causes the water above X to increase in temperature.

....

....

....

....

.... [3]

(ii) Heater Y remains switched off. Explain why the temperature of the water below heater X increases much more slowly than the temperature of the water above heater X.

....

....

....

.... [2]

(b) The hot water tank is covered in a thick layer of insulating material. The material is a plastic that contains a large number of small pockets of trapped air.

Explain why this material is a good insulator.

....

....

....

.... [2]

[Total: 7]

5. M/J/20/22/Q5(b)

Fig. 5.1 shows a microwave oven used to heat soup. The container for the soup is a glass bowl.

Microwaves created inside the oven are reflected by the metal walls.

Fig. 5.1

(i) Explain why the choice of material for the container is important in microwave cooking.

....

....

.... [1]

(ii) The soup is mostly water.

Microwaves are completely absorbed by a few centimetres of water.

As a result, microwaves do not reach the centre of the soup.

The instructions suggest that, after the microwave oven is turned off, the soup:

- is not stirred
- is left for some minutes so that the centre becomes hot.

State the name of and describe each of the **two** processes by which thermal energy transfers throughout the soup after the microwave oven is turned off.

name of first process ....

description ....

....

....

name of second process ....

description ....

....

.... [3]

6. M/J/20/21/Q4

Glass and iron are both conductors of heat. However, glass is a poor conductor of heat and iron is a good conductor of heat.

(a) Describe, using ideas about particles, how the conduction of heat takes place in glass and in iron. You should make it clear why iron is a better conductor of heat.

conduction in glass ....

....

....

....

....

conduction in iron ....

....

....

....

....
[4]

(b) Fig. 4.1 shows apparatus used to show expansion.

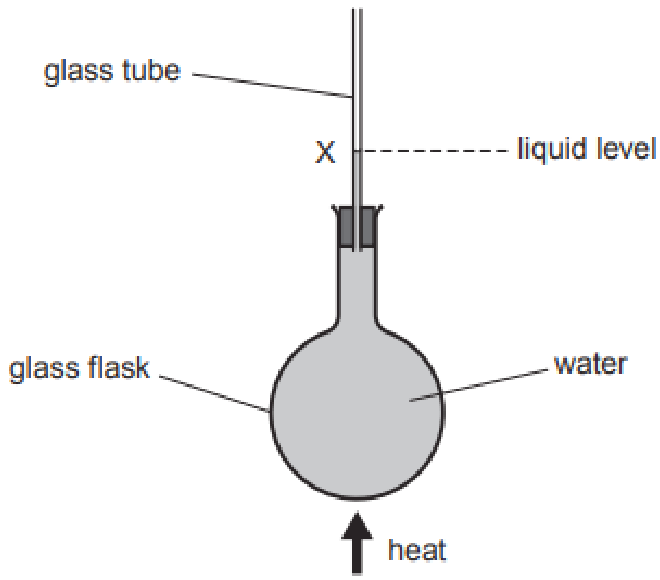


Fig. 4.1

The glass flask, full of water, is heated. A student is surprised when the liquid level X in the glass tube falls for a few seconds before it rises.

(i) Suggest why the liquid level falls and why it then rises.

....

....

....

....

.... [2]

(ii) Describe how heat is transferred throughout the water in the glass flask.

....

....

....

....

.... [2]

7. O/N/19/21/Q9

Fig. 9.1 shows the structure of a water cooler that is used to supply cold water to the workers in a hot office.

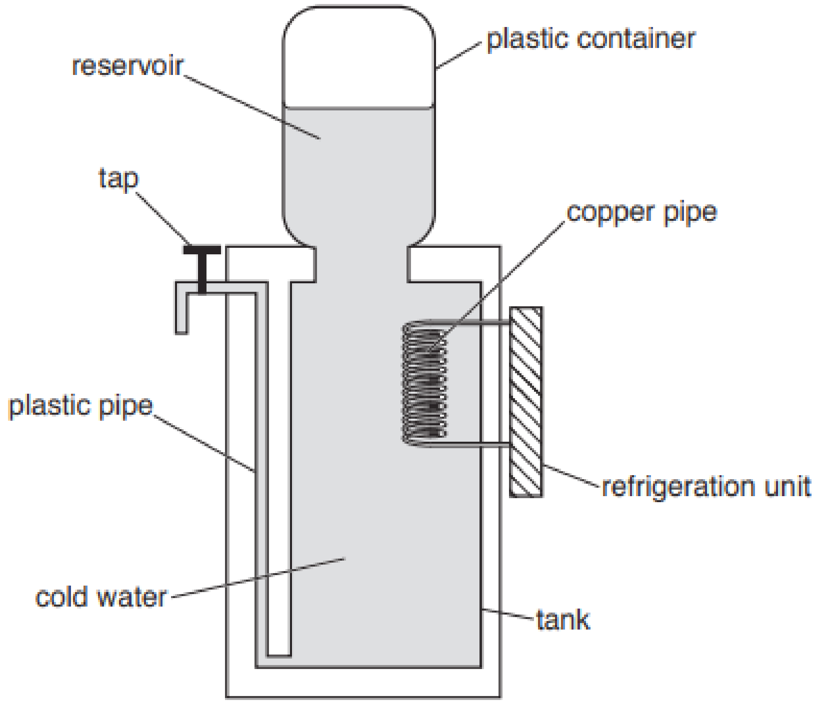


Fig 9.1

(a) The reservoir of water in the plastic container is at room temperature. It does not mix with the cold water in the tank below.

Explain why.

....

....

.... [2]

(b) When the tap is opened, water at room temperature from the reservoir flows down into the tank. Cold water from the tank flows through the plastic pipe and out of the tap.

Cold liquid from the refrigeration unit is pumped through the copper pipe and thermal energy passes through the copper to this liquid.

(i) Suggest why this pipe is made from copper.

....

.... [1]

(ii) Explain, in terms of free electrons, how thermal energy is transferred through the copper.

....

....

....

.... [3]

(iii) As the water near the copper pipe cools, it begins to mix with the rest of the water in the tank.

Explain how this happens.

....

....

....

....

.... [3]

(c) The tap is opened and water at 21 °C flows from the reservoir into the tank.

The specific heat capacity of water is 4200 J / (kg °C).

Calculate the thermal energy that is removed from 250 g of this water to reduce its temperature to 7 °C.

energy = ... [3]

(d) (i) Describe the motion of the molecules in a liquid which is at a uniform temperature throughout.

....

....

.... [2]

(ii) State what happens to the motion of the molecules in a liquid as the temperature of the liquid decreases.

....

.... [1]

8. M/J/19/22/Q10

Fig. 10.1 is a diagram of a soldering iron. Solder is a mixture of metals used to make a permanent contact between electrical wires.

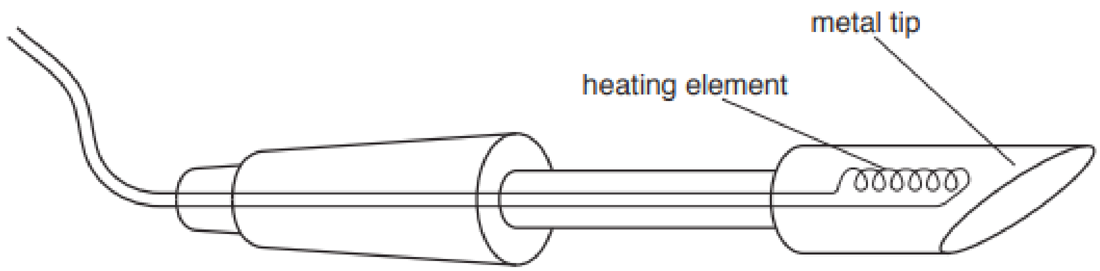


Fig. 10.1

The heating element raises the temperature of the metal tip. When solder is placed against the tip, the solder melts over the wires to be joined. When the solder cools, it solidifies and the permanent connection is made.

The working temperature of the metal tip is 380 °C.

(a) The boxes in Fig. 10.2 show two materials and some different melting points.

material	**melting point**
	1000 °C
metal of the metal tip	380 °C
	200 °C
	20 °C
solder	0 °C

Fig. 10.2

On Fig. 10.2, draw a line from the metal of the metal tip and a line from the solder to a suitable melting point for each. [2]

(b) The heating element is rated at 24 V, 3.3 A.

The heating element is switched on.

The temperature of the metal tip rises from 20 °C to 320 °C in the first 10 s.

(i) Calculate the electrical energy supplied to the heating element in the first 10 s.

energy = ... [2]

(ii) The metal tip is made of copper and has a mass of 2.3 g. The specific heat capacity of copper is 0.39 J / (g °C).

Calculate the thermal energy (heat) gained by the metal tip in the first 10 s.

thermal energy = ... [3]

(c) **(i)** Describe, in terms of free electrons, the process by which heat transfers through the metal tip.

....

....

....

.... [2]

(ii) Heat is lost from the metal tip by convection in the air.

Explain how convection occurs in the air.

....

....

....

.... [2]

9. O/N/18/22/Q3

Fig. 3.1 shows a cylindrical copper kettle that contains cold water.

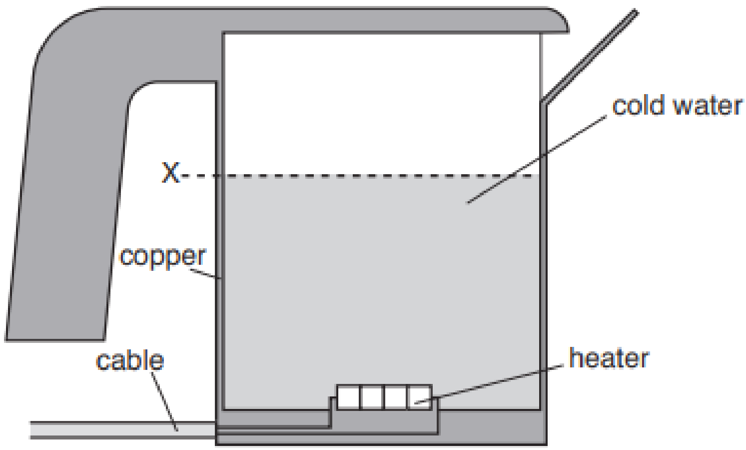


Fig. 3.1

The kettle is used to heat water and there is an electric heater at the base.

(a) State and explain the advantage of heating the water from below.

....

....

....

.... [2]

(b) As the water is heated, it expands.

(i) Explain, in terms of molecules, why water expands when it is heated.

....

....

....

.... [2]

(ii) Copper also expands when heated.

State what happens to level X of the water in the kettle. Explain your answer in terms of the expansion of the copper and the water.

....

....

.... [1]

10. O/N/18/22/Q4

A house has several solar panels on the roof.

These panels use energy from the Sun both to generate electricity and to raise the temperature of water that passes through tubes inside the panels.

(a) The panels on the roof of the house have a black surface.

(i) State how energy from the Sun travels through space before it reaches the Earth.

....

.... [1]

(ii) Explain the advantage of using panels that have a black surface.

....

....

....

.... [2]

(b) On one occasion, the panels are supplying an electric current of 15 A at a voltage of 24 V.

(i) Calculate the electrical energy generated by the panels in one hour.

electrical energy = ... [2]

(ii) In the same time, 51 kg of cold water is pumped through the panels. The temperature of the water increases from 16 °C to 45 °C.

The specific heat capacity of water is 4200 J / (kg °C).

Calculate the increase in thermal energy of the water.

thermal energy increase = ... [3]

11. M/J/17/21/Q3

Fig. 3.1 shows a metal coffee cup on a metal warming plate.

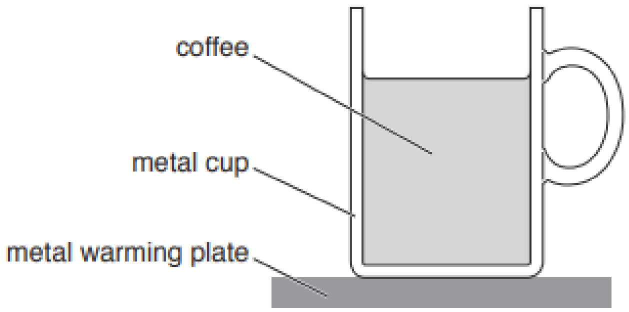


Fig. 3.1

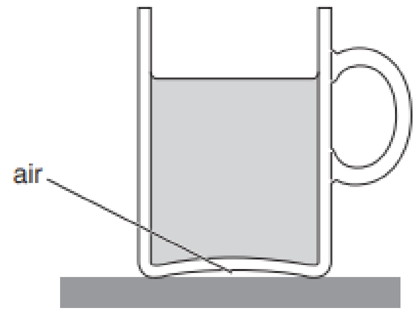


Fig. 3.2

There is a small electrical heater inside the warming plate that keeps the plate hotter than the coffee.

(a) Describe how heat is transferred through the metal and then to all of the liquid in the cup.

....

....

....

....

....

....

.... [3]

(b) A cup of a different shape is placed on the same heater, as shown in Fig. 3.2. The two cups are made of the same metal and contain the same amount of coffee.

Explain why the coffee in the cup in Fig. 3.2 is not kept as warm as the coffee in the cup in Fig. 3.1.

....

.... [1]

(c) The outside surface of the cup can be either black or white and can be either dull or shiny.

(i) Underline which colour **and** which type of surface is best to keep the coffee warm.

black **white** **dull** **shiny** [1]

(ii) Explain your answer to **(c)(i)**.

....

....

[1]

12. O/N/15/21/Q2

Fig. 2.1 shows a freezer for keeping food frozen.

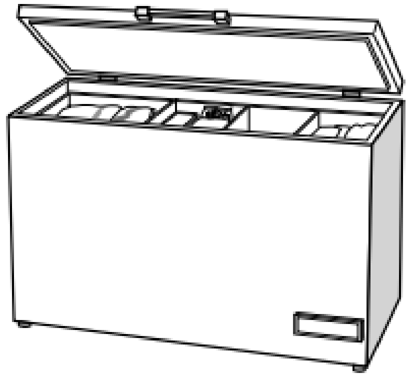


Fig. 2.1

(a) The outside surface of a freezer is painted white.

Explain one advantage of having the outside surface of the freezer painted white.

....

....

.... [2]

(b) (i) The lid of the freezer is closed and air at room temperature is trapped inside the freezer. The freezer is switched on.

State and explain, using ideas about molecules, what happens to the pressure of the air in the freezer, as it cools.

....

....

....

....

.... [3]

(ii) When the freezer reaches its operating temperature, it is more difficult to open the lid.

Explain why.

....

....

.... [1]

13. O/N/15/22/Q4

A copper saucepan with a wooden handle contains cold water. The saucepan is placed on a red-hot heating element that is a part of an electric cooker.

Fig. 4.1 shows the saucepan on the heating element.

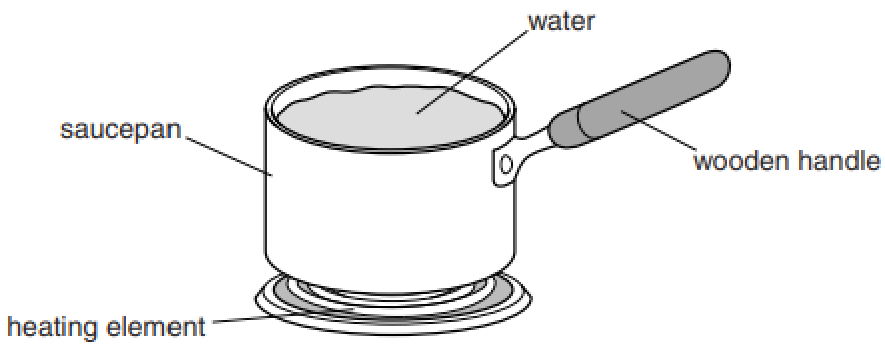


Fig. 4.1

(a) Explain why wood is a suitable material for the handle of the saucepan.

.... [1]

(b) (i) Describe and explain, in terms of free electrons, how thermal energy is transferred through the copper base of the saucepan.

....

....

....

....

....

.... [3]

(ii) Describe and explain how thermal energy is transferred upwards through the water.

....

....

....

....

....

....

.... [3]

14. O/N/14/21/Q10

The casing of an electric kettle is made of white plastic. Fig. 10.1 shows the heating element positioned in the base of the kettle.

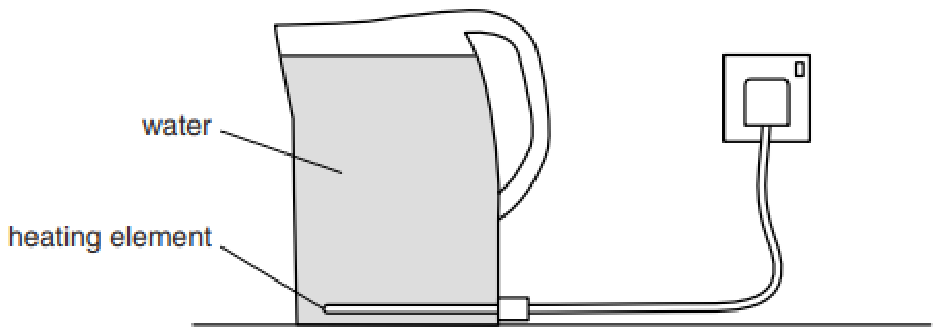


Fig. 10.1

(a) **(i)** The heating element supplies thermal energy to the water at the bottom of the kettle.

Describe and explain how the thermal energy is transferred throughout the water.

....

....

....

....[3]

(ii) Explain why a kettle with its heating element in the water at the top of the kettle does not heat the water uniformly.

....

....[1]

(b) The kettle is powered by a 230V supply. It is switched on for 3.5 minutes and there is a current of 9.6A in the heating element.

(i) Calculate the thermal energy produced in the heating element in this time.

thermal energy = ...[2]

(ii) The kettle contains 1.6 kg of water that was at an initial temperature of 22 °C. The specific heat capacity of water is 4200 J / (kg °C).

Calculate the maximum possible temperature of the water.

temperature = ...[3]

(iii) Suggest one reason why the temperature of the water, after 3.5 minutes, is less than the value calculated in **(b)(ii)**.

....

....[1]

(c) Explain one advantage of

(i) using plastic for the casing of a kettle,

....

....[1]

(ii) choosing white as the colour for the outside of the casing.

....

....[1]

(d) The kettle is switched on again and the water reaches its boiling point. It starts to boil and the kettle remains switched on.

(i) State the meaning of *boiling point*.

....

....[1]

(ii) Explain, in terms of molecules, what happens to the thermal energy that is supplied when the water is boiling.

....

....

....[2]

15. M/J/14/22/Q10

Two metal saucepans contain the same mass of hot water at the same initial temperature.
Pan A is white and pan B is black, but otherwise the two saucepans are identical.
Both saucepans are uncovered and cool under the same conditions.
The cooling curves for the two saucepans are shown in Fig. 10.1.

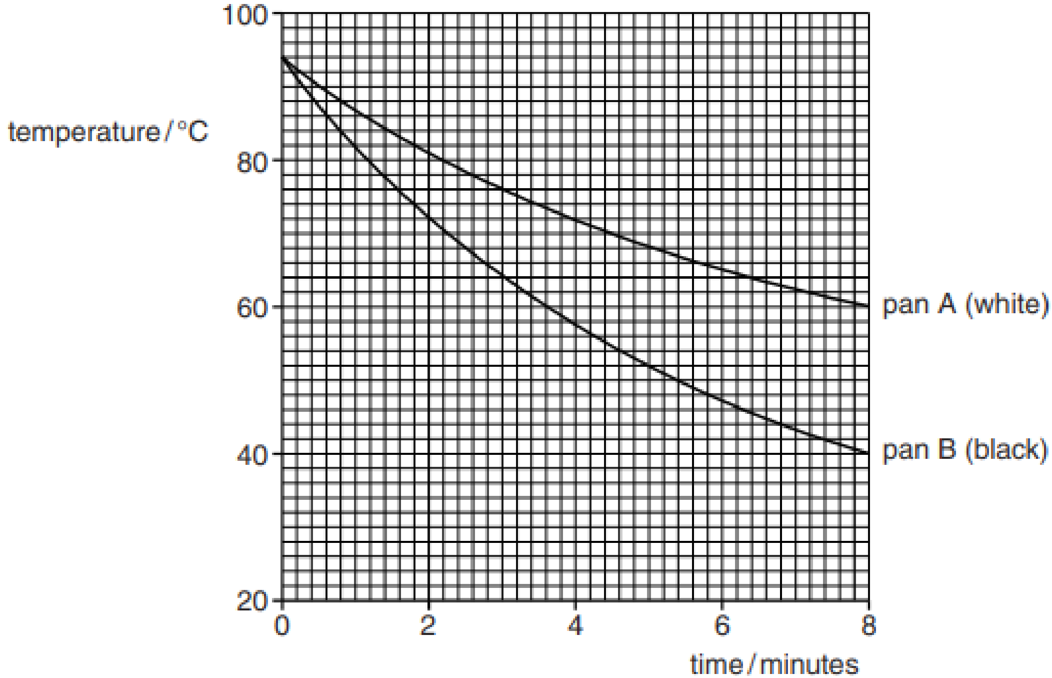


Fig. 10.1

(a) Describe how the water in a pan loses heat by

(i) conduction,

....

....

.... [2]

(ii) convection.

....

....

.... [2]

(b) **(i)** Explain why pan B cools faster than pan A.

....

.... [1]

(ii) Describe and explain how Fig. 10.1 is different when the pans are covered and the experiment is repeated.

....

....

.... [2]

(c) The specific heat capacity of water is 4200 J/(kg °C).

(i) Explain what is meant by *specific heat capacity*.

....

....

.... [2]

(ii) The specific heat capacity of water is very high. Suggest one disadvantage of this when water is used for cooking.

....

.... [1]

(iii) The water in pan A cools for 8 minutes, as shown in Fig. 10.1. During this time, the water loses an average of 9000 J of thermal energy per minute.

1. Calculate the mass of water in pan A.

mass = ... [3]

2. The mass of water in pan B is the same as that in pan A.

Calculate the thermal energy lost from the water in pan B during the 8 minutes.

loss of thermal energy = ... [2]

16. O/N/13/21/Q4

Fig. 4.1 shows food being cooked in an electric grill.

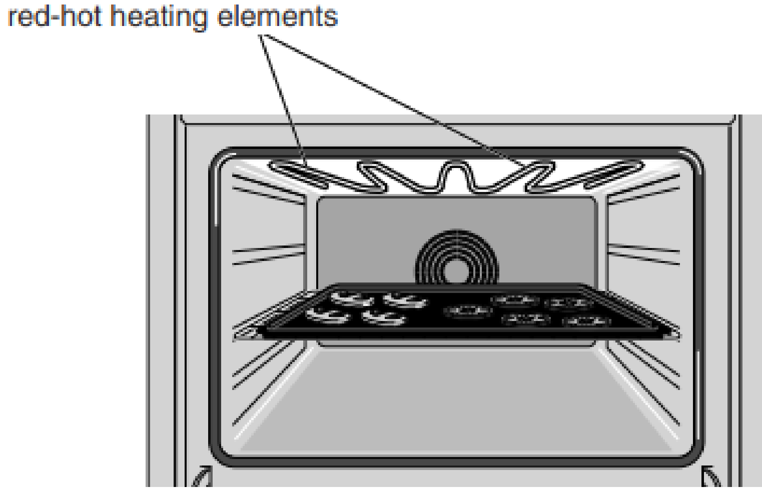


Fig. 4.1

There are red-hot heating elements above the food and thermal energy (heat) is transmitted to the food by radiation.

(a) Explain what is meant, in this case, by *radiation*.

....

....

.... [2]

(b) Explain why very little thermal energy is transmitted to the food by

(i) conduction,

....

.... [1]

(ii) convection.

....

.... [1]

17. O/N/12/21/Q10

Fig. 10.1 shows a domestic heater that is used to heat a room. It contains 16 large bricks each of mass 7.5 kg.

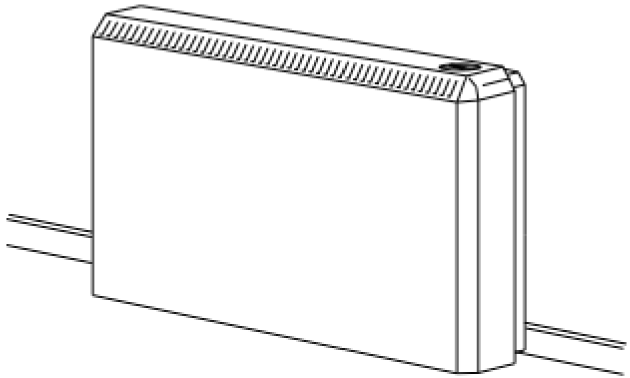


Fig. 10.1

During the night, the bricks are heated from a temperature of 17 °C to 96 °C. The bricks are made from a material that has a specific heat capacity of 2300 J / (kg °C).

(a) Calculate the thermal energy (heat) supplied to the bricks.

thermal energy = ... [3]

(b) During the day, the bricks gradually cool and the stored thermal energy is released to the room. After 7.0 hours, the bricks have cooled to 17 °C.

(i) Calculate the average rate of release of thermal energy to the room during these 7.0 hours.

rate = ... [2]

(ii) At the beginning of the day, the heater releases thermal energy at a greater rate than later in the day.

Suggest why.

....

....

.... [2]

(c) Fig. 10.2 is a diagram of the heater standing on the floor of the room and placed next to one of the walls.

Fig. 10.2

Describe, and explain in detail, how the thermal energy is transferred throughout the room. You may draw on Fig. 10.2.

....

....

....

....

....

.... [5]

(d) State one method of thermal insulation that is used to keep a room warm, and explain why it is effective.

....

....

....

....

.... [3]

18. O/N/12/22/Q5

(a) No thermal energy (heat) is transferred from the surface of the Sun to the Earth by either conduction or convection.

Explain why this is so.

....

....

.... [2]

(b) In a certain country, the climate is very sunny and hot during the day and extremely cold during the night.

Explain how painting the houses white helps to maintain a comfortable temperature both during the day and during the night.

during the day: ....

....

....

during the night: ....

....

....

[3]

19. M/J/12/22/Q3

Fig. 3.1 shows the plan of a bedroom and part of the main room of a house. Other rooms are not shown.

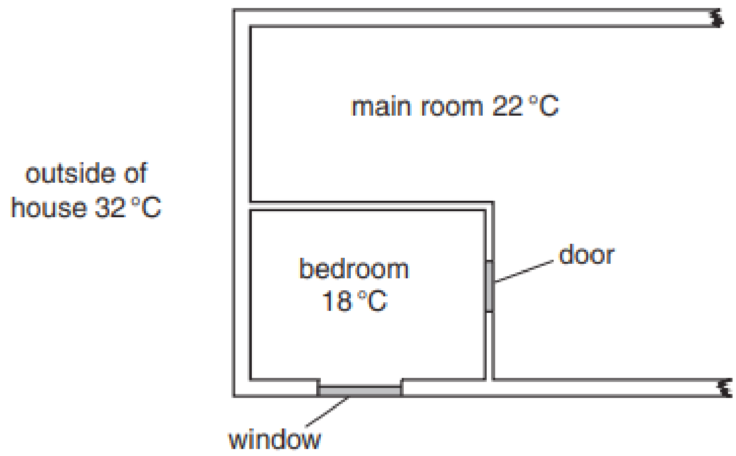


Fig. 3.1

The temperatures of the main room, the bedroom and the outside of the house are shown on Fig. 3.1.

Fig. 3.2 shows all the thermal energy (heat) inputs to the bedroom in one hour.

thermal energy input to bedroom	
through door and walls from main room	50 000 J
through walls from outside of house	2 000 000 J
through window	1 000 000 J
from person sleeping in bedroom	250 000 J

Fig. 3.2

(a) Suggest why more thermal energy enters the bedroom from the outside of the house than from the main room.

....[1]

(b) An air conditioner keeps the temperature constant in the bedroom by removing thermal energy.

(i) Calculate the total thermal energy that the air conditioner removes from the bedroom in one hour.

thermal energy = ...[1]

(ii) The electrical power input to the air conditioner is 300 W.
Calculate the electrical energy input into the air conditioner in 1 hour.

energy = ...[2]

(c) The air conditioner cools the air at the top of the room. This causes a convection current in the room.

Explain how the cold air gives rise to the convection current.

....

....

....[3]

20. M/J/10/22/Q3

Fig. 3.1 shows a metal roof. One side is facing the Sun.

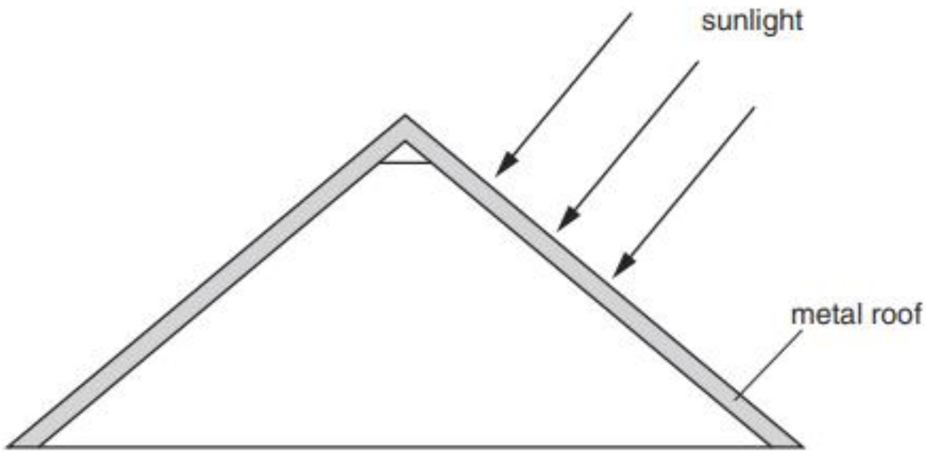


Fig. 3.1

(a) State the means by which thermal energy (heat) is transferred from the Sun to the Earth and explain why other means of thermal energy transfer are not involved.

....

....[2]

(b) Describe how thermal energy is transferred through the metal roof from the heated surface.

....

....

....[2]

(c) During the night, the metal roof loses 1.2×10^6 J of thermal energy and its temperature falls by 20 °C.

The specific heat capacity of the metal in the roof is 400 J/(kg °C).

Calculate the mass of metal in the roof.

mass = ...[2]

21. M/J/09/Q2

Fig. 2.1 shows a metal pan containing water on a cooker. The hotplate heats the water.

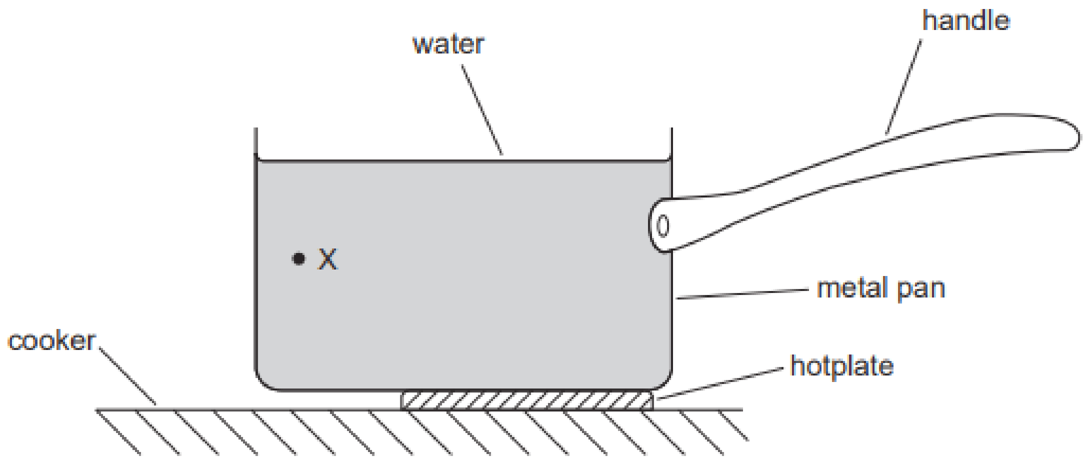


Fig. 2.1

(a) **(i)** State the method of heat transfer through the metal pan.

.... [1]

(ii) Describe how the molecules transfer heat through the metal pan.

....

....

.... [1]

(b) **(i)** On Fig. 2.1, draw an arrow to show the direction of movement of the water at point X. [1]

(ii) Explain why the water moves in this direction.

....

....

....

....

....

.... [3]

22. M/J/08/Q3

During a marathon race, the runner shown in Fig. 3.1 is very hot.

Fig. 3.1

At the end of the race, evaporation and convection cool the runner.

(a) **(i)** Explain how evaporation helps the runner to lose energy. Use ideas about molecules in your answer.

....

....

....

.... [2]

(ii) Explain why hot air rises around the runner at the end of the race.

....

....

....

.... [1]

(b) At the end of the race, the runner is given a shiny foil blanket, as shown in Fig. 3.2. Wearing the blanket stops the runner from cooling too quickly.

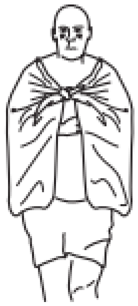


Fig. 3.2

Explain how the shiny foil blanket helps to reduce energy losses.

Use ideas about conduction, convection and radiation in your answer.

....

....

....

....

....

....

....

.... [3]

23. O/N/07/Q4

Fig. 4.1 shows equipment placed on top of a house that uses solar energy to produce hot water.

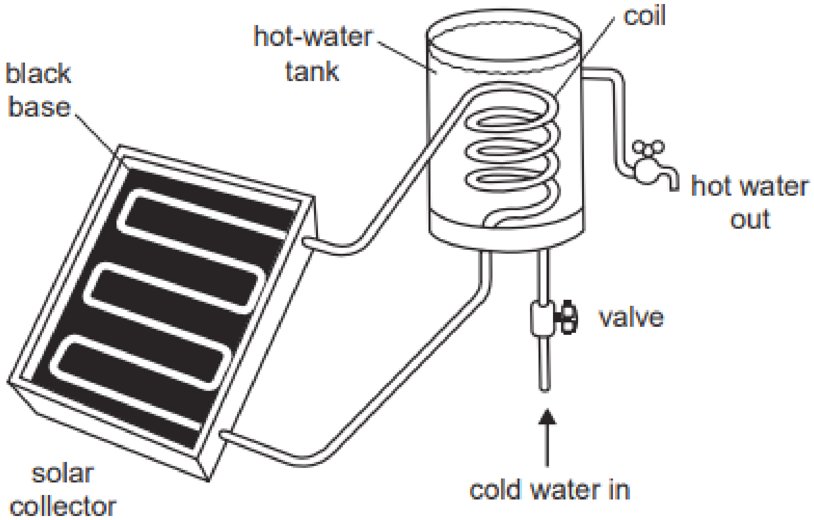


Fig. 4.1

(a) Explain why the solar collector has a black base.

....

....

.... [2]

(b) State and explain why the hot water in the solar collector travels to the hot-water tank.

....

....

.... [2]

(c) Fig. 4.1 does not show any insulation.

(i) Explain why it is important to insulate the hot-water tank.

....

.... [1]

(ii) Explain how the hot-water tank is insulated.

....

.... [1]

24. O/N/06/Q3

Three horizontal rods are placed with one end just above a Bunsen flame. The other end of each rod is coated with wax, as shown in Fig. 3.1.

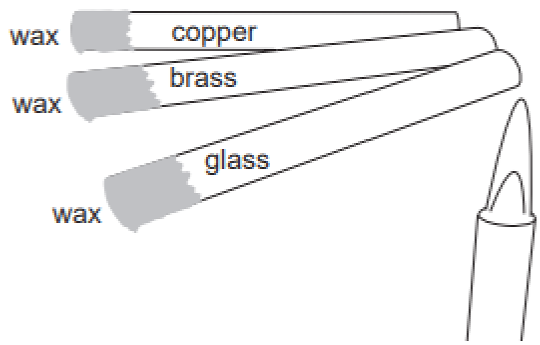


Fig. 3.1

(a) Describe how you would use the apparatus to discover which rod is the best conductor of heat.

....

....

....

....

.... [2]

(b) Two metal teapots are identical except that one is black on the outside and the other is white on the outside, as shown in Fig. 3.2.

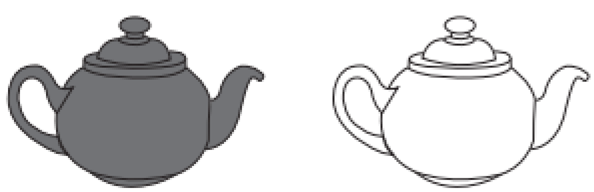


Fig. 3.2

The teapots each contain the same amount of hot water.

State and explain which teapot will cool down more quickly.

....

....

....

....

....

.... [3]

25. M/J/04/Q2

Heat is transferred by conduction, convection and radiation.

(a) **(i)** State which of the three methods is responsible for the transfer of heat from the Sun to the Earth.

....

(ii) Explain why the other two methods cannot be involved in this transfer.

....

....

....
[2]

(b) A hand feels hot when placed above a lighted match, as shown in Fig. 2.1.
Explain in detail how convection causes this to happen.

....

....

....

....

....

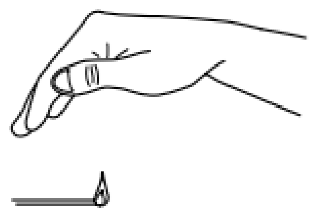


Fig. 2.1 [2]

(c) Fig. 2.2 shows a layer of fibreglass placed between the ceiling of a room and the roof of a house.

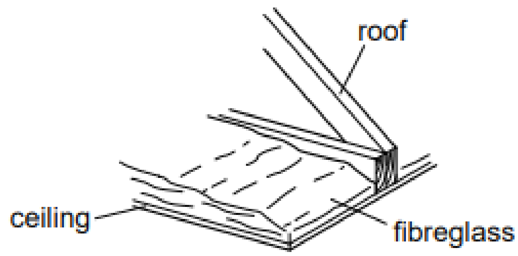


Fig. 2.2

Explain how the layer of fibreglass helps to keep the room warm when it is cold outside.

....

....

....

....[2]

ANSWERS

1.

Question	Answer	Marks
3(a)	particles move apart **or** distance between particles increases	**B1**
	particles able to move further against the forces (of attraction) **or** forces (of attraction) less effective against faster moving particles **or** particles collide harder (with neighbours)	**B1**
3(b)(i)	conduction	**B1**
3(b)(ii)	<u>density</u> of heated water decreases	**B1**
	less dense / heated water rises	**B1**
	cold water sinks / convection current established	**B1**
3(b)(iii)	100 °C **or** 373 K	**B1**
3(b)(iv)	(ΔT =) 100 – 17 **or** 83 (°C)	**C1**
	(ΔE =) $mc\Delta T$ **or** 2.5 × 4200 × (100 – 17) **or** 2.5 × 4200 × 83	**C1**
	8.7×10^5 (J)	**A1**

2.

Question	Answer	Marks
3(a)	chemical potential energy **and** no other energy	**B1**
3(b)(i)	radiation **or** infrared mentioned	**B1**
	radiation **and** infrared (travel to the man)	**B1**
	energy / infrared / radiation <u>absorbed</u> by the man / pullover **or** travels in straight line **or** travels at the speed of light **or** increases (man's) internal energy	**B1**
3(b)(ii)	(shiny surface) <u>reflects</u> radiation (to man)	**B1**
	less (thermal) energy escapes **or** more (thermal) energy reaches / absorbed by man	**B1**
3(b)(iii)	**EITHER** black (surface) is good absorber / poor reflector of infrared radiation	**B1**
	more (thermal) energy absorbed **or** more thermal energy transferred to man	**B1**
	OR trapped air **and** reduced convection / air poor conductor / air is a good insulator	**(B1)**
	less (thermal) energy lost	**(B1)**

3.

Question	Answer	Marks
2(a)	dull black line 100	**B1**
	dull white 60 **and** shiny white 40	**B1**
2(b)	hot air rises	**B1**
	density of hot air less (than cold air) **or** air expands on heating	**B1**
	hot air rises <u>and</u> <u>cold</u> air falls **or** a complete current described	**B1**

4.

Question	Answer	Marks
3(a)(i)	water expands (as it is heated) **or** molecules move apart	**B1**
	density (of heated water) decreases **or** (heated water) rises	**B1**
	it rises **and** the cold water falls **or** convection current established	**B1**
3(a)(ii)	water (below X) heated by conduction **or** no convection occurs	**C1**
	water is a poor conductor	**A1**
3(b)	air / plastic is a poor / slow (thermal) conductor	**B1**
	trapping the air prevents convection (in air) **or** no <u>free</u> electrons (in plastic)	**B1**

5.

5(b)(i)	(microwaves) must pass through **or** not be absorbed (by container) **or** not reflected	**B1**
5(b)(ii)	convection **and/or** conduction named	**B1**
	(convection involves) hot liquid rises (cold liquid falls)	**B1**
	(conduction involves) energy passes from molecule to molecule	**B1**

6.

Question	Answer	Marks
4(a)	***glass*** molecules vibrate	**B1**
	molecules pass on energy / vibration from one to another	**B1**
	contains <u>many / more</u> free electrons / electrons that can move	**B1**
	electrons collide with atoms / molecules (and gain / lose energy)	**B1**
4(b)(i)	glass warms up (before water) **or** glass expands first	**B1**
	water expands more than glass	**B1**
4(b)(ii)	convection **or** hot water rises	**B1**
	convection <u>current</u> formed **or** hot water rises **and** cold water falls **or** density of water falls (as it heats)	**B1**

7.

Question	Answer	Marks
9(a)	it is less dense (than the cooler water)	**B1**
	it floats on the cooler water / cooler water cannot move up / cooler water remains on the bottom	**B1**
9(b)(i)	copper / metal is a good (thermal) conductor	**B1**
9(b)(ii)	vibrating atoms / ions / particles / molecules **or** electrons gain energy	**B1**
	atoms / ions / particles / molecules hit the electrons **or** electrons travel (a long distance through the copper)	**B1**
	electrons hit / transfer energy to (distant) atoms / ions / particles / molecules	**B1**

9(b)(iii)	it contracts **or** its density increases	**B1**
	it sinks	**B1**
	less dense / warmer water rises **or** sets up a convection current	**B1**
9(c)	14(°C) **or** 0.25 (kg) **or** 250 / 1000 (kg) seen	**C1**
	(Q =) $mc\Delta t$ **or** $0.25 \times 4200 \times (21 - 7)$ **or** $0.25 \times 4200 \times 14$	**C1**
	14 700 J **or** 15 000 J	**A1**
9(d)(i)	any **two** from: molecules / they move in clusters	**B1**
	slide over each other molecules / they move throughout the liquid	**B1**
9(d)(ii)	average speed / <u>kinetic</u> energy decreases	**B1**

8.

Question	Answer	Marks
10(a)	*metal tip* 1000 °C	**B1**
	solder 200 °C	**B1**
10(b)(i)	(E=)VIt algebraic or numerical	**C1**
	790 J	**A1**
10(b)(ii)	(Q=) mcT algebraic or numerical	**C1**
	$2.3 \times 300 \times 0.39$	**C1**
	270 J	**A1**
10(c)(i)	electrons <u>move</u> around **or** electrons diffuse through (metal)	**M1**
	pass on energy to other parts / hit atoms or other electrons / (transfer energy) in all directions	**A1**
10(c)(ii)	hot air / heated air rises	**B1**
	(hot air) less dense (than cold air)	**B1**
10(d)(i)	two different metals	**B1**
	complete circuit with meter in series **and** one junction labelled H on hot tip	**B1**
10(d)(ii)	ANY TWO from • can measure rapid temperature changes • can measure high temperatures • is small in size • has a small heat capacity • can give electrical signal / be read by computer	**B1 B1**

9.

Question	Answer	Marks
3(a)	all the water is heated **or** the water is mixed up **or** water heated uniformly **or** distributes heat (better)	**B1**
	heated water rises **or** cold water sinks **or** convection transfers thermal energy (upwards)	**B1**
3(b)(i)	molecules move / vibrate fast<u>er</u> / <u>more</u> kinetic energy	**B1**
	molecules push each other apart **or** molecules move apart **or** space between molecules increases **or** vibrate with greater amplitude	**B1**
3(b)(ii)	rises **and** liquids expand more (than solids)	**B1**

10.

Question	Answer	Marks
4(a)(i)	radiation **or** infrared (radiation / waves) **or** light	**B1**
4(a)(ii)	it / a black surface is a good absorber / poor reflector of radiation	**B1**
	<u>more</u> energy / power output **or** <u>more</u> electricity produced	**B1**
4(b)(i)	(P =) VIt **or** 24 × 15 × 3600 **or** 24 × 15 × 60 **or** 22 000 (J)	**C1**
	1.3×10^6 J	**A1**
4(b)(ii)	(ΔQ =) $mc\Delta T$ **or** 29 (°C) **or** 45 – 16 (°C)	**C1**
	51 × 4200 × 29 **or** 51 × 4200 × (45 – 16)	**C1**
	6.2×10^6 J	**A1**

11.

Question	Answer	Marks
3(a)	conduction in metal **or** convection (in liquid) mentioned by name	**B1**
	conduction explained as heat / energy passing from molecule to molecule **or** movement / diffusion / collision of (free) electrons	**B1**
	convection explained by rising of hot liquid **or** correct density changes	**B1**
3(b)	air is a bad conductor **or** less area in contact / all of cup does not touch plate	**B1**
3(c)	white **and** shiny	**B1**
	less radiation emitted / less emission	**B1**

12.

(a) poor absorber / good reflector of (infra-red) radiation B1
(**not** with poor emitter)
less thermal energy absorbed B1

(b) (i) (pressure / it) decreases B1
molecules slow down B1
less frequent / less violent (molecular) collisions **with wall** B1

(ii) (pressure difference causes) a downward force on lid
or pressure outside > pressure inside B1

13.

(a) wood is a poor / not a conductor **or** (good) insulator (of heat) B1

(b) (i) vibrating atoms / ions / particles / molecules **or** electrons gain energy B1
atoms / ions / particles / molecules hit free electrons **or** electrons travel (a long distance through the copper / saucepan) B1
electrons hit / transfer energy to (distant) atoms / ions / molecules / particles B1

(ii) hot / heated water expands / is less dense B1
hot / heated water / less dense water rises B1
(sets up) circulation / convection (current) **or** cold water sinks B1

14.

(a)	**(i)**	**heated / hot water** expands **or** density of **heated / hot water** decreases	B1
		(heated / hot water) rises	B1
		convection (current) / circulation set up **or** (heated / hot water) rises **and** cold water sinks	B1
	(ii)	convection transfers heat upwards **or** less dense / heated / hot water (already) at top	B1
(b)	**(i)**	(Q =) VIt **or** $230 \times 9.6 \times 3.5$ **or** $230 \times 9.6 \times 3.5 \times 60$ **or** 7728	C1
		$4.6(368) \times 10^5$ J	A1
	(ii)	(ΔT =) Q / mc **or** $4.6(3680) \times 10^5 / 1.6 \times 4200$	C1
		69 (°C)	C1
		91 °C	A1
	(iii)	evaporation **or** thermal energy / heat in **plastic casing / element / surroundings** (i.e. air or environment)	B1
(c)	**(i)**	poor conductor (heat or electricity) **or** less heat lost / cooler to touch **or** less risk of shock	B1
	(ii)	poor emitter **and** less heat lost / of radiation / IR (**not** poor absorber)	B1
(d)	**(i)**	**temperature** where liquid and vapour/gas coexist **or** where liquid (**not** substance) boils (at atmospheric pressure)(**allow** becomes vapour/gas)	B1
	(ii)	(work done) against / overcoming forces between molecules **or** molecules gain P.E. (**ignore** K.E. increases)	B1
		changes to P.E. / molecules separate	B1

15.

(a) (i) (conduction occurs) through or in metal / pan **or** from water to metal / pan
or molecules vibrate **or** molecules collide
or (free) electrons (in metal) move [B1]

vibration / energy / heat passed **from molecule to molecule** clear
or energy passed on by electrons colliding (with atoms/molecules or electrons) [B1]

(ii) hot air or air over water rises or hot water rises [B1]
hot air or hot water expands **or** hot air or water less dense [B1]

(b) (i) black objects radiate heat more (than white) [B1]

(ii) (both) graphs higher (after start)
or temperature falls less (in same time) / slower
or takes longer to cool [B1]

less **evaporation** occurs **or** less **convection** [B1]

(c) (i) heat / energy to change the temperature by 1 °C / unit temp [C1]

heat / energy to change the temperature of 1 kg / unit mass by 1 °C / unit temp [A1]

(ii) long time to warm / boil water / cook
or scalds / burns when touched
or more energy needed (to warm water) [B1]

(iii) **1.** 34(°C) **or** 94–60 seen [C1]
(m=) $Q / c\Delta T$ algebraic or numerical with any clear Q or ΔT [C1]
0.5(042) kg [A1]
2. $0.50 \times 4200 \times 54$ [C1]
110 000 **or** 114(353) J [A1]

16.

(a) (energy transmitted) by electromagnetic/infra-red (wave)/can travel
through a vacuum B1
infra-red **or** visible < λ < microwaves **or** λ just longer than visible
(i.e. infra-red scores 2/2) B1

(b) (i) air is a poor conductor B1

(ii) convection occurs (primarily) upwards/hot air rises (**not** heat rises) B1

17.

(a)	16×7.5 **or** 120 **or** 96–17 **or** 79	C1
	$(Q =)mc\Delta T$ **or** $120 \times 2300 \times 79$	C1
	$2.2(2.1804) \times 10^7$ J	A1
(b) (i)	$2.2 \times 10^7/7$ **or** $2.2 \times 10^7/(7 \times 60)$ **or** $2.2 \times 10^7/(7 \times 3600)$	C1
	3.1×10^6 J/h or 5.2×10^4 J/min or 870 J/s or W	A1
(ii)	(heater/bricks) hot(ter) (**not** room cooler)	B1
	great(er) temperature difference (between heater and room)	B1
(c)	air (next to heater) gets hot or conduction through metal/casing	B1
	expands or radiation or IR (radiation)	B1
	less dense	B1
	rises	B1
	circulation or convection current or arrows on Fig. 10.2	B1

(d)	double glazing/cavity walls/ceiling tiles/carpet/curtains/loft insulation etc.	**or**	shiny foil	B1
	traps air		radiation reflected	M1
	air is poor conductor/convection prevented		IR radiation/ back into room	A1

18.

(a)	space is a vacuum/empty	B1
	these methods need matter/medium/molecules **or** do not occur in vacuum	B1

(b) any **three** of:

day: white is a poor absorber/good reflector

day: less heat absorbed/less heating (of house)

night: white is a poor emitter/radiator

night: less heat emitted/heat loss (from house)

anywhere: of IR/radiation/radiant heat B3

19.
(a) large(r) temperature difference (between bedroom and outside) OR outside is hot(ter than main room) B1

(b) (i) 3 300 000 J(/hour) B1

(ii) (E =) $P \times t$ in any form; 300 × 60 × 60 C1
1.08 × 106 J; 1.1 × 106 J
OR 0.3 **kWh** A1

(c) cold air sinks B1

(cold air has a) high(er) density or contracts B1

hot air rises
OR hot air has a low(er) density
OR (hot) air comes in to replace cold air B1

20.
(a) radiation or infra-red or electromagnetic waves B1
travels through space/vacuum or does not require medium/molecules/particles or medium required for conduction and/or convection or for other methods B1

(b) conduction occurs
or atoms/particles/molecules vibrate
or electrons given energy B1
heat/energy/vibration passed on from one particle to another
or electrons move to other parts/diffuse/hit atoms B1

(c) (Q =) mcT algebraic or numerical in any form (e.g. $1.2 \times 10^6 = m \times 400 \times 20$) C1
150 kg A1

21.
(a) (i) conduction B1

(ii) molecules hit each other or molecules pass vibration on
or free electrons move (through metal) and hit molecules B1

(b) (i) downwards at or near X B1

(ii) hot water less dense or cold water more dense B1
hot water rises (not heat rises) or cold water falls B1
convection current mentioned or water flows to replace hot water that rises
or rising and falling described or water cools at surface B1

22.

(a) **(i)** **molecules/atoms/particles** escape/leave **or** liquid **molecules** change to gas/vapour B1
fastest/high energy molecules evaporate/energy needed to break bonds/latent heat B1

(ii) hot air less dense **or** cold air more dense **or** air expands **or** body heat **conducted** into air B1

(b) trapped air
air is a bad conductor/good insulator
convection **current** reduced **or** (air) **flow** reduced
(shiny) heat/IR/radiation reflected **or shiny** less radiation/heat emitted
evaporation reduced/air more humid, etc. ANY 3 lines 1 each B3

23.

(a) good absorber (**not** good absorber and emitter) (**not** attracts) B1
radiation **or** infra red (**not** heat) B1

(b) hot water rises (**not** heat rises) B1
by convection (currents) **or** density explanation B1

(c) **(i)** reduce/avoid/prevent loss of heat B1

(ii) cover/wrap in lagging/any sensible material (**not** wood/insulation, **acc.** plastic tank) B1

24.

(a) time **or** observe when wax melts/falls **or** states first to melt/fall B1
first to do so **or** less wax left (after given time) (transfers heat best) B1

(b) black **or** black cools quickly **M**1
better emitter (of heat) A1 **OR** better radiator/black radiates white doesn't A1
radiation/infra-red A1 of heat/infra-red A1
Accept in terms of white teapot (NOT better emitter and absorber/conductor)

25.

(a) **(i)** radiation B1
(ii) no molecules or medium (to vibrate, conduct, convect) / vacuum B1

(b) **hot air** rises B1
(hot) air expands / density decreases B1

(c) fiberglass or air is a bad conductor/ insulator / lags / reduces heat flow
fiberglass traps air or prevents convection B1
(ignore radiation statements) B1